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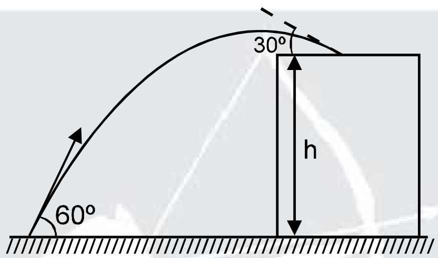
PHYSICS

TARGET : JEE-ADVANCED-2024

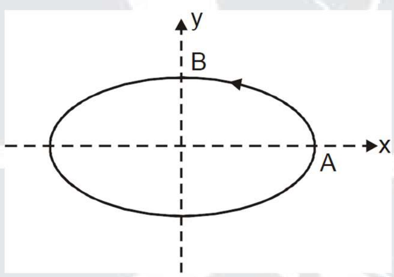
CAPS-13^{+}

1. A stone projected at an angle of 60º from the ground level strikes at an angle of 30º on the roof of a building of height 'h'. Then the speed of projection of the stone is :



(A) $\sqrt{2gh}$ (B) $\sqrt{6gh}$ (C) $\sqrt{3gh}$ (D) $\sqrt{gh}$

2. A particle is moving along an elliptical path with constant speed. As it moves from A to B, magnitude of its acceleration :

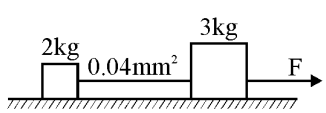


(A) continuously increases
(B) continuously decreases
(C) Remains constant
(D) first increases and then decreases

3. A particle moves along the parabolic path $y = ax^2$ in such a way that the y-component of the velocity remains constant, say c. The x and y coordinates are in meters. Then acceleration of the particle at x = 1 m is

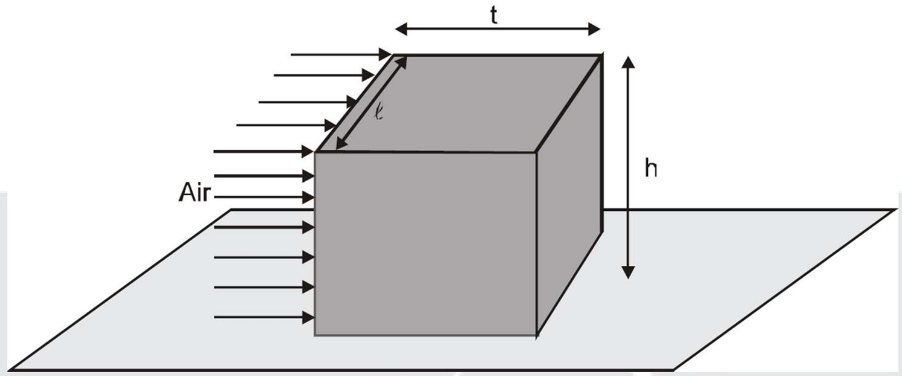
(A) $ac\hat{k}$ (B) $2ac^2\,\hat{j}$ (C) $-\frac{c^2}{4a^2}\hat{i}$ (D) $-\frac{c}{2a}\hat{i}$

4. Two bodies of masses 2kg and 3kg are connected by a metal wire of cross section 0.04 mm^2. Breaking stress of metal wire is 2.5 GPa. The maximum force F that can be applied to 3kg block so that wire does not break is : Neglect friction



(A) 100N (B) 150 N (C) 200 N (D) 250 N

5. A block of dimensions $\ell \times t \times h$ and uniform density ρ_w rests on a rough floor. Wind blowing with speed V and of density ρ_a falls perpendicularly on one face of dimension $\ell \times h$ of the block as shown in figure. Assuming that air is stopped when it strikes the wall and there is sufficient friction on the ground so that the block does not slide, the minimum speed V so that the block topples is :



(A) $\left(\frac{\rho_w g}{\rho_a h}\right)^{1/2}.t$ (B) $\left(\frac{\rho_a g}{\rho_w h}\right)^{1/2}.t$ (C) $\left(\frac{g}{h}\right)^{1/2}.t$ (D) None of these

6. Two particles A and B are initially 40 m apart. A is behind B. Particle A is moving with uniform velocity of 10 m/s towards B. Particle B starts from rest moving away from A with constant acceleration of 2 m/s^2. The minimum distance between the two is :

(A) 15 m (B) 20 m (C) 25 m (D) 30 m

7. A uniform disc of radius R lies in the x - y plane with its centre at origin. Its moment of inertia about z-axis is equal to its moment of inertia about an axis along the line $y = K_1x + K_2$ where K_1 and K_2 are constants. Then choose the option (s) having possible values of K_1 and K_2.

(A) $K_1 = 1, K_2 = +\frac{R}{\sqrt{2}}$ (B) $K_1 = 1, K_2 = -\frac{R}{2}$

(C) $K_1 = -1, K_2 = +\frac{R}{2}$ (D) $K_1 = -1, K_2 = -\frac{R}{\sqrt{2}}$

8. A particle moves along positive branch of the curve $y = \frac{x}{2}$, where $x = \frac{t^3}{3}$. {x is in meter t in sec}. Then :

(A) The velocity of the particle at t = 1 sec. is $(\hat{i} + \frac{1}{2}\hat{j})$ m/s

(B) The acceleration of the particle at t = 1 sec. is $(2\hat{i} + \hat{j})$ m/s^2

(C) The particle is speeding up at t = 1 sec.

(D) The particle is slowing down at t = 1 sec.

9. Consider a spring that exerts the following restoring force :

$$F = \begin{cases} -kx & \text{for } x > 0 \\ -2kx & \text{for } x < 0 \end{cases}$$

A mass m on a frictionless surface is attached to this spring, displaced to x = A by stretching spring and released.

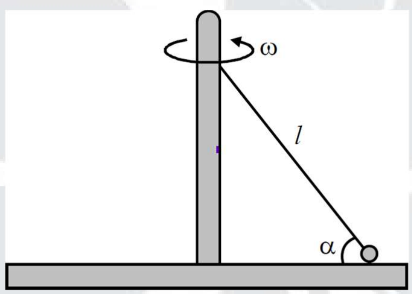
(A) The time period of motion is $\pi\sqrt{\frac{m}{k}}\left[1+\frac{1}{\sqrt{2}}\right]$

(B) The time period of motion $2\pi\sqrt{\frac{m}{k}}\left[1+\frac{1}{\sqrt{2}}\right]$

(C) The most negative value of x that the mass m reaches is $\left(-\frac{A}{2}\right)$

(D) The most negative value of x that the mass m reaches is $-\left(\frac{A}{\sqrt{2}}\right)$

10. A circular platform rotates around a vertical axis with angular velocity ω = 10 rad/sec. On the platform, there is a ball of mass 1 kg, attached to the long axis of the platform by a thin rod of length ℓ = 10 cm (α = 30º). Select correct alternative(s).



(A) Tension is rod is 10N

(B) Normal force on ball is 5N

(C) Net force on ball is zero

(D) Net electromagnetic force on ball is zero

11. In a race on a track of length ℓ, there is a tie between two cars each taking time t. Initial velocities and constant accelerations of cars are v_1, v_2 and a_1, a_2. Which of the options are correct

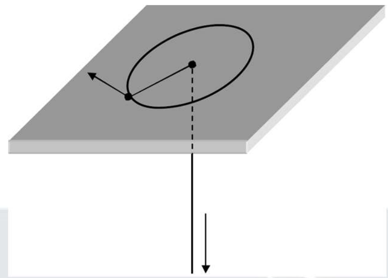
(A) $t = \frac{2(v_2 - v_1)}{(a_1 - a_2)}$

(B) $\ell = \frac{2(v_1 - v_2)(v_1a_2 - v_2a_1)}{(a_1 - a_2)^2}$

(C) $t = \frac{2(v_1 - v_2)}{(a_1 - a_2)}$

(D) $\ell = \frac{2(v_1 - v_2)(v_1a_1 - v_2a_1)}{(a_1 - a_2)^2}$

12. A mass is attached to the end of a string. The mass moves on a frictionless table , the string passes through a hole in the table, under which some one is pulling on the string to make it taut at all times. Initially, the mass moves in a circle of radius R with kinetic energy E_0. The string is slowly pulled, until the radius of the circle is halved. Choose the correct option(s)



(A) The tension in string is $\frac{2E_0}{R}$

(B) The tension in string is $\frac{L^2}{mR^3}$. Where L is angular momentum of the particle about center.

(C) Work done to reduce the radius half is $\frac{3L^2}{2mR^2}$ Where L is angular momentum about center.

(D) Work done to reduce the radius half is $3E_0$

13. Nuclei of radioactive element A are being produced at a constant rate α. The element has a decay constant λ. At time t = 0, there are N_0 nuclei of the element. Which of the following statements are correct:

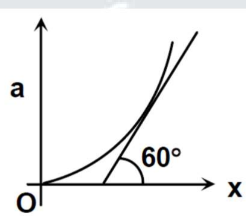
(A) The number of nuclei of A at time t is $\frac{1}{\lambda}$ $(\alpha - (\alpha - \lambda N_0)e^{-\lambda t})$

(B) The number of nuclei of A at time t is $\frac{1}{\lambda}$ $(\alpha - \lambda N_0)e^{-\lambda t})$

(C) If $\alpha = 2N_0\lambda$, the number of nuclei of A after one half life of A is $\frac{3}{2}N_0$

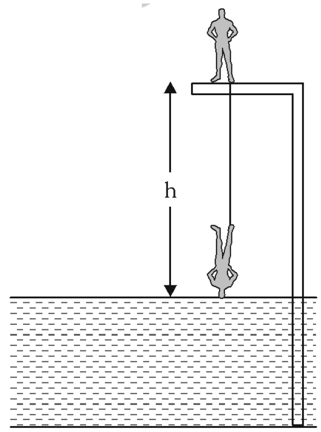
(D) The number of nuclei of A after a very long time is $3N_0$

14. A particle starts moving with initial velocity 3 m/s along x-axis from origin. Its acceleration is varying with x according to equation $a = kx^2$. Figure shows a – x graph. At x = $\sqrt{3}$ m tangent to the graph makes an angle 60° with positive x-axis as shown in diagram. Then at x = $\sqrt{3}$



(A) $v = \sqrt{(\sqrt{3}+9)}$ m/s (B) a = 1.5 m/s^2 (C) v = $\sqrt{12}$ m/s (D) a = 3 m/s^2

15. A man of height h_0 = 2m and mass m = 50 kg is bungee jumping from a platform situated at a height h = 25 m above a lake. One end of an elastic rope is attached to his foot and the other end is fixed to the platform. He starts falling from rest in vertical position. The length and elastic properties of the rope are chosen so that his speed will have been reduced to zero at instant when his head reaches the surface of water. Ultimately the jumper is hanging from the rope with his head 8m above the water, (take g = 10 m/s^2) (air resistance is very–very small, center of mass of man at mid–point)



(A) The maximum acceleration achieved during the jump is approximately 40 m/s^2.

(B) Natural length of elastic rope is 13m.

(C) When finally man comes to at rest, elongation in the rope is 2m.

(D) Total work done by elastic rope is 250 J.

16. A ball of small size is rolled off the edge of a horizontal table. The ball has an initial speed v_o and lands on the floor at some distance from the base of the table. Which of the following statements concerning the fall of the ball is/are **TRUE**?

(A) The time of flight depends on the height of the table.

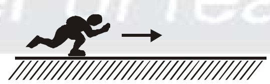
(B) The horizontal component of average velocity for the duration of flight will be v_0.

(C) The ball will accelerate for the duration of flight.

(D) The ball will have a longer flight time if v_0 is increased.

PARAGRAPH

If a man has a velocity varying with time given as v = 3t^2 , v is in m/s and t in sec then :



17. Find out the velocity of the man after 3 sec.

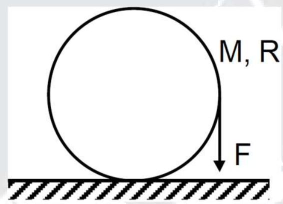
(A) 18 m/s (B) 9 m/s (C) 27 m/s (D) 36 m/s

18. Find out his displacement after 2 seconds of his start :

(A) 10 m (B) 6 m (C) 12 m (D) 8 m

19. Two men A and B, A standing at periphery of circle of radius R and B standing at centre of the circle. A starts moving on the periphery with constant acceleration and B starts moving with constant velocity v at an angle θ, from the line joining initial position of A and centre, such that they meet during subsequent motion, for this purpose maximum possible value of v is $\sqrt{\frac{aR}{\alpha\theta}}$. Here α is an integer. Find α.

20. A circular disc of mass M is placed over a rough horizontal surface. Coefficient of friction is μ. A force F is applied on the periphery of the disc, tangentially in vertically downward direction. The disc will undergo pure rolling only if $F \leq F_0$. When $F = F_0$ value of $\frac{(2-3\mu)F_0}{\mu Mg}$ is :



ANSWER KEY

1.	(C)	**2.**	(B)	**3.**	(C)	**4.**	(D)	**5.**	(A)
6.	(A)	**7.**	(AD)	**8.**	(ABC)	**9.**	(AD)	**10.**	(AB)
11.	(AB)	**12.**	(ABCD)	**13.**	(AC)	**14.**	(AB)	**15.**	(ABC)
16.	(ABC)	**17.**	(C)	**18.**	(D)	**19.**	2	**20.**	3